

FINAL REPORT

1. INTRODUCTION

The project "Smart Agricultural Production Optimization Engine" aims to develop an advanced software system that utilizes data-driven insights to optimize agricultural production for different crops. By integrating key environmental factors such as Nitrogen (N), Phosphorous (P), Potassium (K) levels, soil temperature, humidity, pH, rainfall, and crop types, OptiCrop seeks to provide intelligent recommendations to farmers for maximizing yields and resource efficiency. The knowledge and insights gained through the OptiCrop project can be utilized by agricultural researchers, policymakers, and stakeholders to advance the understanding of crop-environment interactions and inform the development of evidence-based agricultural strategies. The OptiCrop project revolutionizes agricultural practices by providing farmers with a smart production optimization engine. By integrating key environmental factors and crop-specific recommendations, OptiCrop enables farmers to achieve optimal yields, improve resource efficiency, and contribute to sustainable agricultural practices.

Scenario 1: Smart Crop Recommendation for Farmers

A farmer enters soil and environmental details such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH level, and rainfall. The system analyzes the data and recommends the most suitable crop for maximum yield and better farming efficiency.

Scenario 2: Crop Suitability and Environmental Assessment

A user wants to understand whether the current soil and climate conditions are suitable for a particular crop. The application evaluates the input parameters and provides insights about crop compatibility, environmental suitability, and productivity potential.

Scenario 3: Agricultural Research and Policy Planning

An agricultural researcher or policymaker uses the system to analyze crop-environment relationships and identify patterns in agricultural production. The platform helps in making data-driven decisions for sustainable farming strategies and resource optimization.

2. IDEATION PHASE

2.1 Problem Statement

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - who are they?	Describe the customer and their attributes here
I'm trying to	List their outcome or "job" the care about - what are they trying to achieve?	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way - what bothers them most?	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists - what needs to be solved?	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view - how does it impact them emotionally?	Describe the emotions the result from experiencing the problems or barriers

Reference: <https://miro.com/templates/customer-problem-statement/>

Example:



Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a small-to-medium scale farmer	decide which crop to grow each season for the best possible yield	I can't reliably tell which crop suits my soil's N, P, K, pH levels and the	I lack access to an easy, data-driven tool that analyzes my soil and climate readings and	uncertain, financially exposed, and reliant on guesswork instead of evidence

			season's temperature, humidity, and rainfall	recommends the most suitable crop	
PS-2	a farmer or agronomist who already has a crop in mind	confirm whether my current soil and climate conditions are actually suitable for that crop	I have no quick way to check my soil and climate parameters against the crop's ideal requirements before investing time and money	there's no accessible system that evaluates my inputs against known crop-environment compatibility data	anxious about wasting resources on a crop that may underperform or fail
PS-3	an agricultural researcher or policymaker	analyze patterns in crop-environment relationships across regions to plan sustainable farming strategies	I lack a consolidated platform that aggregates soil, climate, and crop production data for analysis	existing agricultural data is scattered, manually maintained, and not structured for cross-crop or cross-region comparison	limited in my ability to make timely, evidence-based recommendations for sustainable agriculture

2.2 Empathy Map Canvas

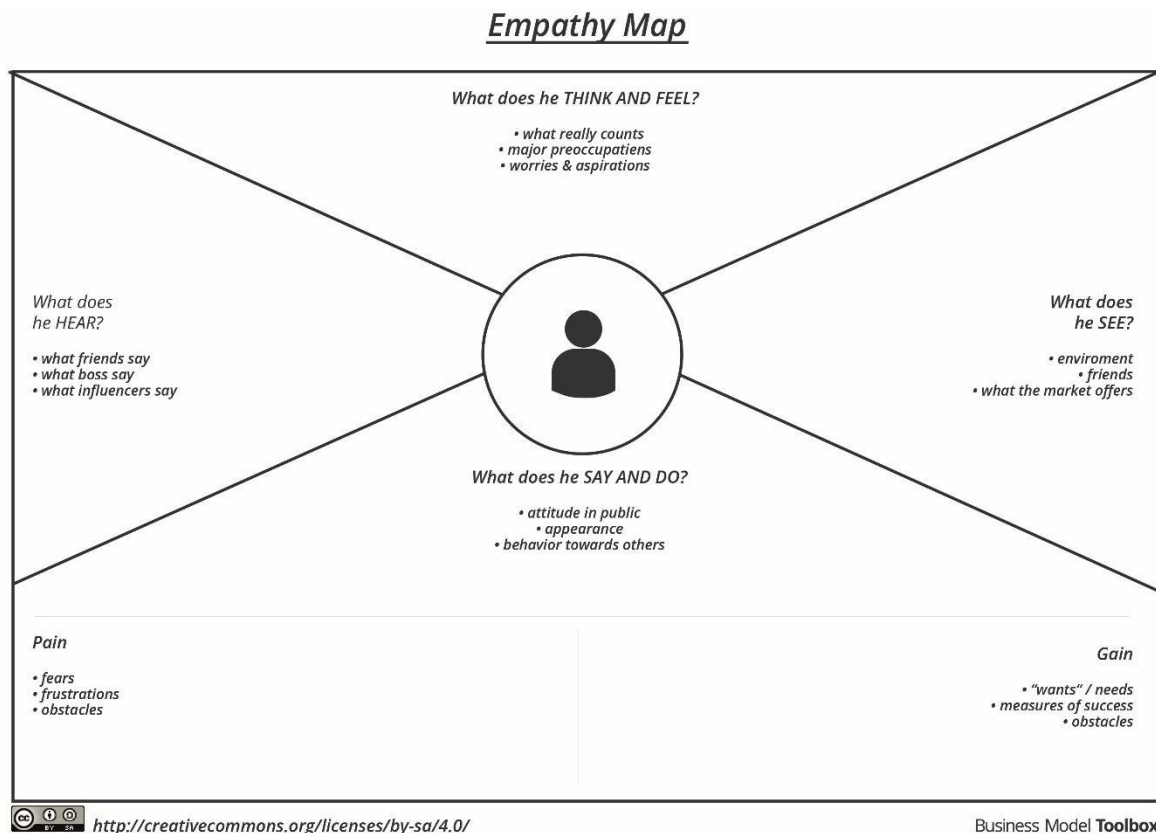
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

It is a useful tool to help teams better understand their users.

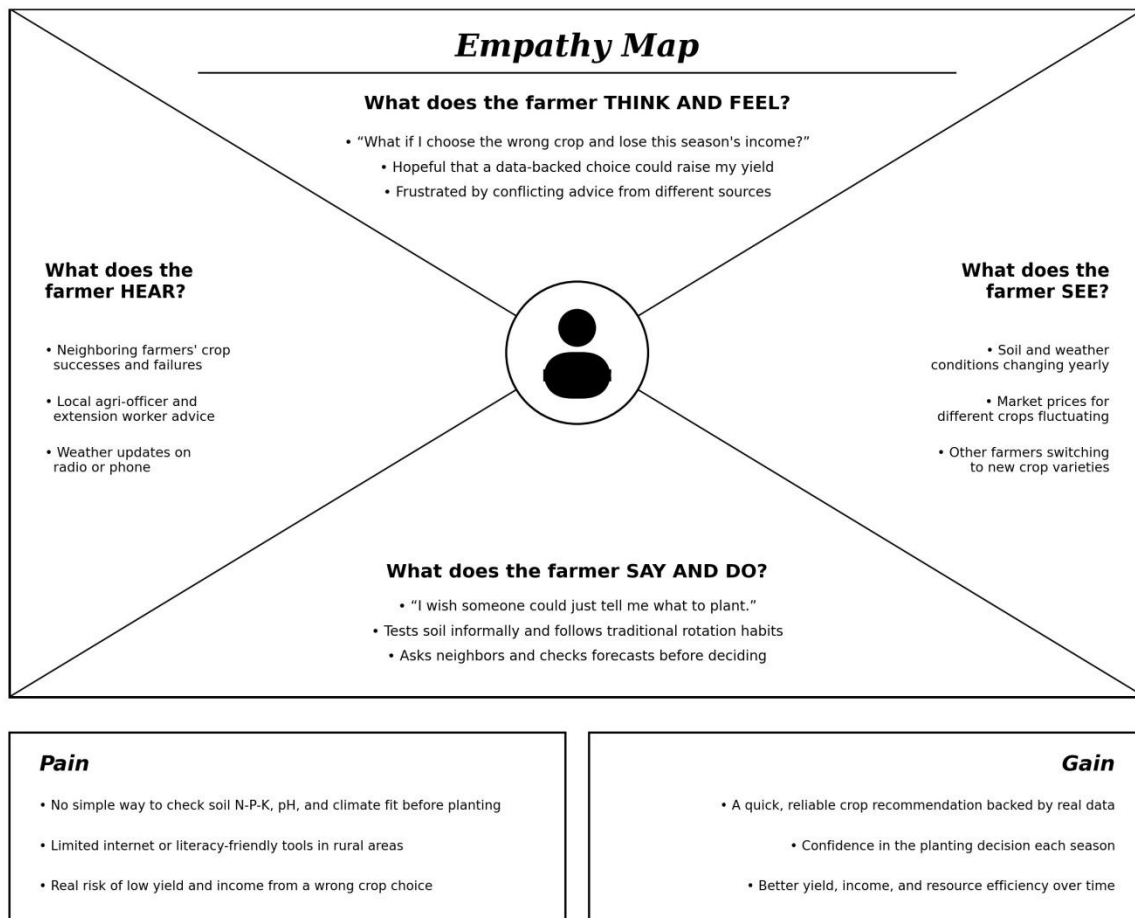
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

Example:



Reference: <https://www.mural.co/templates/empathy-map-canvas>

Empathy Map: OptiCrop — Farmer Persona



2.3 Brainstorming

Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: <https://www.mural.co/templates/brainstorm-and-idea-prioritization>

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Team Gathering & Collaboration:

The team came together to review the day-to-day challenges farmers, agronomists, and policymakers face when deciding which crop to grow under a given set of soil and climate conditions. Through discussion, the team identified a clear opportunity to apply data-driven decision support to agriculture — using soil nutrient levels, weather parameters, and crop history to guide better farming decisions.

Problem Statement:

Farmers often lack access to a reliable, data-driven tool that recommends the most suitable crop based on soil nutrient levels (Nitrogen, Phosphorous, Potassium), pH, temperature, humidity, and rainfall. This leads to suboptimal crop selection, reduced yields, inefficient use of resources, and limited support for researchers and policymakers studying crop-environment relationships. There is a need for an intelligent system — **OptiCrop** — that analyzes environmental and soil parameters to recommend optimal crops, assess crop-environment suitability, and support data-driven agricultural research and policy planning.

Step-2: Brainstorm, Idea Listing and Grouping

Idea Group	Brainstormed Ideas
Data Input & Preprocessing	Collect N, P, K, pH, temperature, humidity & rainfall inputs; validate and clean manual/sensor entries; support CSV and IoT sensor integration; handle missing or outlier values.
Crop Recommendation Engine	ML classification model (e.g. Random Forest, XGBoost, Decision Tree) trained on crop data; rank top-N suitable crops; confidence score per recommendation; explainable feature-importance insights.
Suitability & Environmental Assessment	Compare a chosen crop against ideal parameter ranges; suitability score/percentage; visual indicators (gauge or radar chart); suggested corrective actions (e.g. fertilizer adjustment).
Research & Policy Analytics	Dashboards for regional crop-environment trend analysis; correlation analysis between soil parameters and yield; exportable reports for policymakers; historical trend visualization.
User Experience & Accessibility	Simple web/mobile entry form for farmers; multilingual support; voice or SMS-based input for low-literacy users; offline-first design for rural connectivity.

Step-3: Idea Prioritization

Idea	Impact	Feasibility	Priority
ML-based crop recommendation engine	High	High	P1
Data validation & preprocessing pipeline	High	High	P1
Suitability / environmental assessment scoring	High	Medium	P1
Research / policy analytics dashboard	Medium	Medium	P2
Explainable AI insights (feature importance)	Low	Medium	P3
Multilingual / voice / SMS input support	Medium	Low	P3
IoT sensor integration	Medium	Low	P3

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Soil & Environmental Data Input	Manual entry of N, P, K, temperature, humidity, pH and rainfall values Auto-fetch real-time weather/rainfall data via Weather API Bulk upload of soil-test data via CSV
FR-4	Crop Recommendation Generation	Run ML model on submitted soil & climate parameters Display top recommended crop(s) with confidence score Show suitability score for a specific crop selected by the user
FR-5	Research & Analytics Dashboard	View historical crop-environment trends and patterns Filter data by region, season, or crop type Export analysis reports as PDF / CSV
FR-6	Notifications & Alerts	Email/SMS delivery of crop recommendation report Alert farmer when input conditions are unsuitable for any crop

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

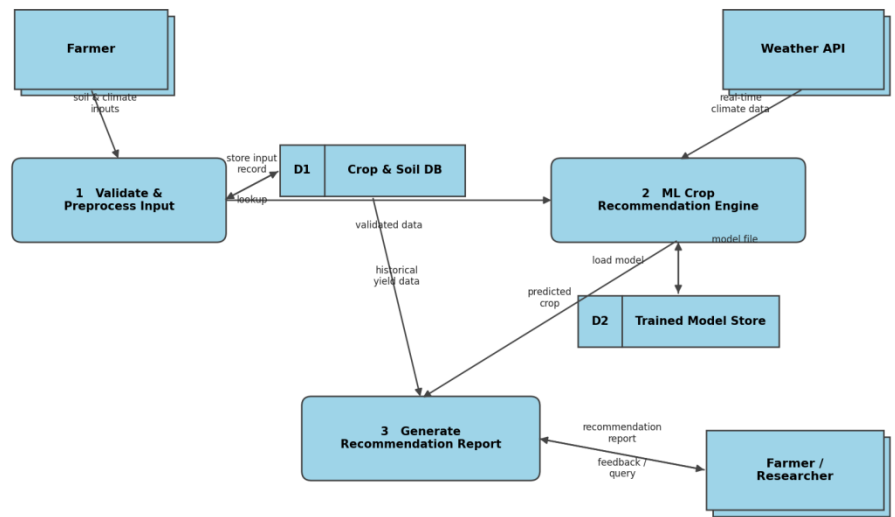
FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, intuitive dashboard that lets farmers with limited technical knowledge enter inputs and understand recommendations easily; supports regional languages.
NFR-2	Security	Encrypted data transmission (HTTPS/SSL), role-based access control, secure storage of user and soil/crop data, input validation against malicious data.
NFR-3	Reliability	Consistent and accurate crop recommendations with proper error handling for invalid or missing input values; automated model validation before deployment.
NFR-4	Performance	Crop recommendation results should be generated within 2-3 seconds of submitting input parameters, even under concurrent user load.
NFR-5	Availability	System should be available at least 99.5% of the time via cloud-based deployment with monitoring and automated failover.
NFR-6	Scalability	Architecture should support a growing number of farmers/researchers and expanding datasets without degradation in response time.

3.3 Data Flow Diagram

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [OptiCrop Process Flow](#)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Farmer (Mobile / Web User)	Registration	USN-1	As a farmer, I can register for the application using my mobile number, email, and password.	I can access my dashboard after registering.	High	Sprint-1
	Registration	USN-2	As a farmer, I will receive a confirmation OTP/email once I have registered for the application.	I can verify my account via OTP or confirmation email link.	High	Sprint-1
	Login	USN-3	As a farmer, I can log into the application using my registered email/mobile number and password.	I can access my dashboard after successful login.	High	Sprint-1
	Soil & Environmental Data Input	USN-4	As a farmer, I can enter my soil values (N, P, K, pH) and climate values (temperature, humidity, rainfall) manually.	The input form validates and accepts all required fields.	High	Sprint-2
	Soil & Environmental Data Input	USN-5	As a farmer, I can let the app auto-fetch local weather data instead of entering it manually.	Weather data is fetched and pre-filled correctly for my location.	Medium	Sprint-2
	Crop Recommendation Generation	USN-6	As a farmer, I can submit my data and receive the top recommended crop along with a confidence	Recommendation is displayed within a few seconds of submission.	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
			score.			
	Crop Recommendation Generation	USN-7	As a farmer, I can check the suitability score for a specific crop that I choose.	Suitability score and a short explanation are displayed.	Medium	Sprint-3
	Notifications & Alerts	USN-8	As a farmer, I receive an email/SMS with my crop recommendation report.	The report is delivered to my registered email or mobile number.	Low	Sprint-3
Agricultural Researcher / Policymaker	Research & Analytics Dashboard	USN-9	As a researcher, I can view aggregated crop-environment trends across regions and seasons.	Dashboard displays charts that can be filtered by region, season, or crop type.	Medium	Sprint-3
	Research & Analytics Dashboard	USN-10	As a researcher, I can export crop-environment analysis reports in PDF/CSV format.	Report downloads correctly with the selected filters applied.	Low	Sprint-4
Administrator	User Management	USN-11	As an administrator, I can view and manage registered farmer and researcher accounts.	Admin can view, edit, or deactivate user accounts.	Medium	Sprint-4
	Model Management	USN-12	As an administrator, I can upload or update the trained ML model used for crop recommendations.	New model file is validated and deployed without downtime.	Low	Sprint-4

3.4 Technology Stack

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

OptiCrop: System Architecture Diagram

Reference: [Crop Recommendation Dataset — Kaggle \(N, P, K, temperature, humidity, pH, rainfall\)](#)

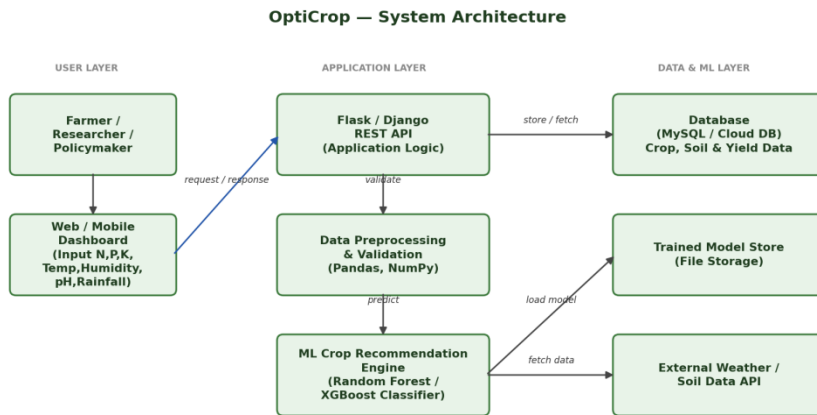


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web/mobile dashboard where farmers, researchers, and policymakers enter soil & climate parameters (N, P, K, temperature, humidity, pH, rainfall) and view crop recommendations.	HTML, CSS, JavaScript, React.js / Flask-Jinja templates
2.	Application Logic-1	Validates and preprocesses user-input soil and environmental data before passing it to the recommendation engine.	Python (Pandas, NumPy)
3.	Application Logic-2	Core ML-based crop recommendation engine that predicts the most suitable crop for the given soil & climate conditions.	Python, Scikit-learn (Random Forest / Decision Tree / XGBoost Classifier)
4.	Application Logic-3	Crop suitability & environmental assessment module that scores compatibility between input conditions and candidate crops.	Python (Flask REST API)
5.	Database	Stores historical soil, climate, crop and yield records used for training and analytics.	MySQL / PostgreSQL
6.	Cloud Database	Centralized, multi-user data store enabling researchers and policymakers to access aggregated crop-	IBM Cloudant / Firebase Firestore / MongoDB Atlas

		environment data.	
7.	File Storage	Stores trained ML model files, datasets, and generated reports.	Local Filesystem / IBM Cloud Object Storage / AWS S3
8.	External API-1	Fetches real-time weather and rainfall data to supplement farmer-provided inputs.	OpenWeatherMap API / IBM Weather API
9.	External API-2	Integrates government soil-health and agricultural advisory data for richer recommendations.	Soil Health Card API / data.gov.in Agriculture API
10.	Machine Learning Model	Classification model trained on N-P-K, temperature, humidity, pH and rainfall data to recommend the optimal crop.	Random Forest Classifier / XGBoost (trained on crop recommendation dataset)
11.	Infrastructure (Server / Cloud)	Application deployment on local system or cloud. Local Server Configuration: Flask development server on localhost Cloud Server Configuration: Containerized deployment with auto-scaling	Local, Docker, Kubernetes, IBM Cloud Foundry / AWS EC2 / Heroku

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used to build the web app, ML pipeline and UI.	Flask / Django, Scikit-learn, Pandas, NumPy, React.js
2.	Security Implementations	Security and access controls implemented to protect user data and the API.	HTTPS/SSL, JWT-based authentication, password hashing (bcrypt), input validation, role-based access (farmer/researcher/admin)
3.	Scalable Architecture	Architecture is designed as a 3-tier system (presentation, application/ML logic, data) so the ML inference service can scale independently.	Microservices via Docker containers, orchestrated with Kubernetes
4.	Availability	Ensures the recommendation service remains available	Load balancer, multiple app instances, database replication,

S.No	Characteristics	Description	Technology
		under load or failure.	cloud auto-scaling
5.	Performance	Design considerations to keep prediction latency low and handle concurrent requests.	Redis caching for frequent queries, optimized model inference, CDN for static assets, asynchronous request handling

4. PROJECT DESIGN

4.1 Problem Solution Fit

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Small & medium-scale farmers Agricultural researchers & scientists Govt. policymakers & agri-stakeholders Rural farming communities across India	6. CUSTOMER CONSTRAINTS CC Limited digital literacy among rural farmers No internet in remote areas Low budget for agri-tech tools Lack of soil testing equipment Language barriers (regional)	5. AVAILABLE SOLUTIONS AS Manual soil testing labs (slow, expensive) Traditional farming knowledge Generic govt. crop guidelines Costly agri-tech apps (limited rural access) Extension officer advice	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Select the right crop for local soil & climate conditions Avoid fertilizer waste (N,P,K) Maximize yield & income Assess crop suitability before planting season Plan farming strategy with data	9. PROBLEM ROOT CAUSE RC No affordable, data-driven tool for individual farm-level crop selection in India Farmers lack access to soil analytics + crop advisory in one integrated platform	7. BEHAVIOUR BE Farmers rely on neighbors or tradition for crop selection Trial-and-error planting cycles Visit local agri officers rarely Researchers use spreadsheets for manual data analysis	
Identify strong TR & EM	3. TRIGGERS TR Poor harvest season Soil test shows nutrient gaps Govt. awareness campaigns Peer farmer success stories	10. YOUR SOLUTION (OptiCrop) SL ML-powered web app (Flask) trained on agricultural data Input: N, P, K, Temperature, Humidity, pH, Rainfall Output: Best crop recommendation + suitability score Visualization dashboard for researchers (Matplotlib/Seaborn) Simple UI — no tech expertise needed by farmers	8.1. CHANNELS — ONLINE CH Flask web application (browser) Mobile-friendly interface Agri-portal integrations	Extract online & offline CH or BE
	4. EMOTIONS: BEFORE / AFTER EM Before: Anxious, uncertain about crop choice & yield outcome After: Confident, informed, resource-efficient farming		8.2. CHANNELS — OFFLINE CH Agri extension officer demos Farmer training workshops Local govt. agri-centers	

4.2 Proposed Solution

Proposed Solution Template: Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers in India lack access to a data-driven tool for selecting the right crop based on their specific soil and environmental conditions. Without scientific guidance on Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall, farmers resort to guesswork, leading to poor yields, resource wastage, and financial losses. Agricultural researchers and policymakers also lack a unified platform to analyze crop-environment patterns for evidence-based decision making.
2.	Idea / Solution description	OptiCrop is an ML-powered web application built using Python (Flask, Scikit-learn) that accepts key soil and environmental parameters as input and recommends the most suitable crop for maximum yield. Key features include: (1) Smart Crop Recommendation Engine – predicts optimal crop from N, P, K, temperature, humidity, pH and rainfall inputs; (2) Crop Suitability Assessment – evaluates whether conditions are suitable for a specific crop; (3) Data Visualization Dashboard – Matplotlib and Seaborn-based charts for researchers and policymakers; (4) User-friendly Flask web interface accessible without technical expertise.
3.	Novelty / Uniqueness	OptiCrop uniquely integrates 7 critical agri-environmental parameters (N, P, K, temperature, humidity, pH, rainfall) into a single ML-based recommendation engine – a combination not available in free, accessible tools for Indian farmers. Unlike generic crop guides, OptiCrop provides farm-specific, real-time predictions. It serves three distinct user groups (farmers, researchers, policymakers) from one platform, and uses Scikit-learn classification models trained on real agricultural datasets to ensure high accuracy.
4.	Social Impact / Customer Satisfaction	OptiCrop directly benefits small and medium-scale farmers by reducing crop failure risk and improving resource efficiency (less fertilizer waste, better water use). For India's ~100 million farming households, even a modest yield improvement translates into significant income gains and food security. Researchers gain a data analysis tool for crop-environment studies, while policymakers can design targeted, evidence-based agricultural schemes. The platform promotes sustainable farming and contributes to UN SDG 2 (Zero Hunger) and SDG 12 (Responsible Production).

5.	Business Model (Revenue Model)	Freemium SaaS Model: Basic crop recommendation is free for farmers via the web app, driving mass adoption. Premium tier (subscription) for agri-businesses, cooperatives, and research institutions offers bulk analysis, API access, and detailed reports. B2G (Business to Government): License the platform to state agriculture departments for integration into farmer advisory schemes (e.g., PM-KISAN). Data Monetization: Aggregated, anonymized crop-environment data sold as insights to agri-input companies and seed manufacturers.
6.	Scalability of the Solution	OptiCrop is built on Python and Flask, which are cloud-deployable on AWS, Azure, or GCP with minimal infrastructure cost. The ML model can be retrained with expanded regional datasets to cover more crops and geographies across South Asia and globally. The modular architecture allows adding new input parameters (e.g., satellite imagery, IoT soil sensors) and new output modules (fertilizer dosage, irrigation schedule). Mobile app development (Android/iOS) will further expand reach to rural farmers with smartphones.

4.3 Solution Architecture

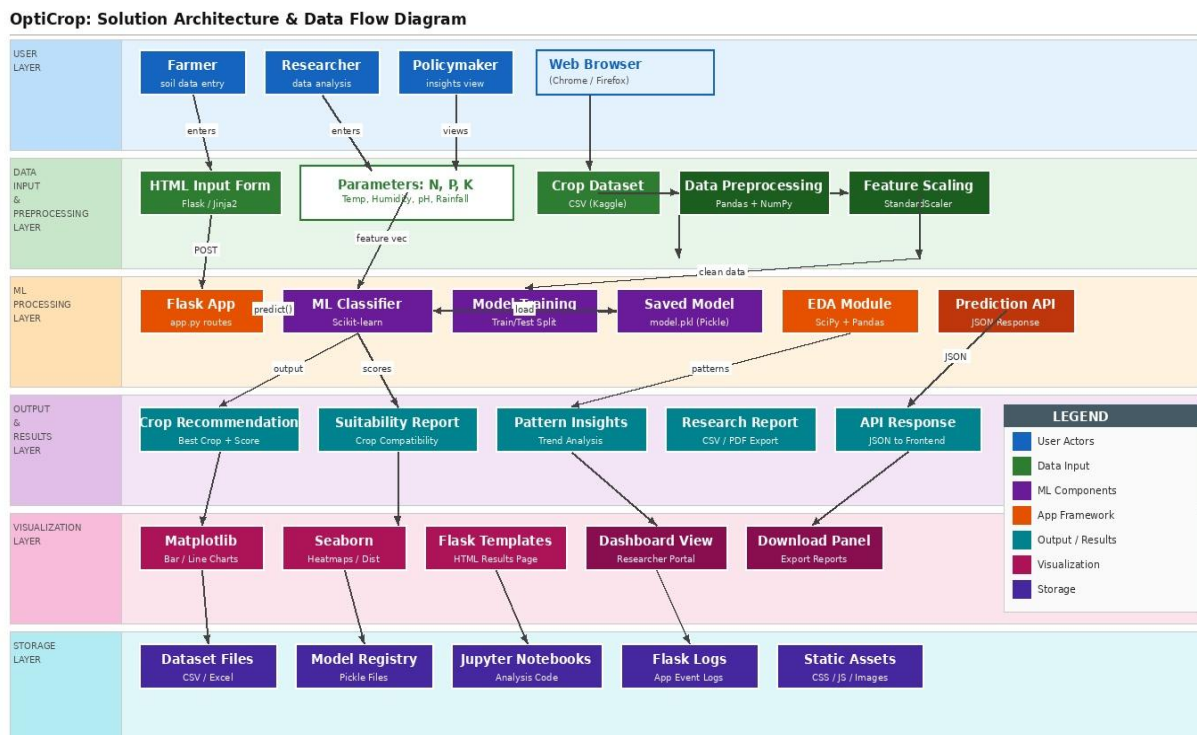


Figure 1: Solution Architecture and Data Flow of OptiCrop - Smart Agricultural Production Optimization Engine

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning logic

A Sprint fixed period or duration in which a team works to complete a set of tasks

An Epic is a **big task or project** that is too large to complete in one sprint. It is broken down into **smaller tasks (stories)** that can be completed over multiple sprints.

A Story is a small task. It is part of an **Epic**.

A Story Point is a number that represents how much effort a story takes to complete. (usually in form of Fibonacci series)

- Very Easy task
- Normal task
- Moderate task
- Difficult task

Storypoint -5 (1,2,3,5)

Sprint 1

Data Collection and Analysis (Epic 2)

Download and collect agricultural dataset (USN1) 2

Import required Python libraries (USN2) 1

Read and explore dataset structure and features (USN3) 2

Perform univariate analysis on N, P, K, temperature, humidity, pH, rainfall (USN4) 3

Conduct bivariate analysis on agricultural parameters vs crop suitability (USN5) 3

Perform multivariate analysis to discover patterns among variables (USN6) 3

Define Problem and Understanding (Epic 1)

Identify and define the agricultural problem for OptiCrop (USN7) 2

Gather and analyze business requirements and project objectives (USN8) 1

Conduct literature survey on crop recommendation techniques (USN9) 3

Analyze social and business impact of AI-driven crop recommendation (USN10) 3

Total Story Point in Sprint 1= 2+1+2+3+3+3+2+1+3+3=23

Sprint 2

Data Pre-Processing (Epic 3)

Check for null values and handle missing data (USN11) 2

Detect and treat outliers affecting model performance (USN12) 2

Extract seasonal crop information and prepare relevant features (USN13) 3

Split dataset into training and testing sets (USN14) 1

Model Building (Epic 4)

Apply K-Means Clustering to identify patterns in agricultural conditions (USN15) 4

Train Logistic Regression model for crop prediction (USN16) 5

Evaluate model performance and compare results (USN17) 3

Save best-performing model and generate crop predictions (USN18) 3

Application Building (Epic 5)

Design and develop HTML pages for interactive user interface (USN19) 5

Build Python backend and integrate with trained ML model (USN20) 5

Run, test, and validate the OptiCrop application (USN21) 3

Total story Point in Sprint 2= 2+2+3+1+4+5+3+3+5+5+3=36

Total Story Points

Sprint 1 = 23

Sprint 2 = 36

Velocity= Total Story Points Completed/ Number of Sprints

Total story Points= 23+36=59

No of Sprints= 2

Velocity =59/2

29.5(Story Points per Sprint)

Your team's velocity is 29.5 Story Points per Sprint.

5.2 Project Planning

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Priority
Sprint-1	Define Problem and Understanding	USN-1	As a researcher, I want to identify and define the agricultural problem that the crop recommendation system aims to solve.	High
Sprint-1		USN-2	As a stakeholder, I want to gather and analyze business requirements to understand project objectives and expected outcomes.	High
Sprint-1		USN-3	As a developer, I want to conduct a literature survey to study existing crop recommendation techniques and machine learning approaches.	Medium
Sprint-1		USN-4	As a policymaker, I want to analyze the social and business impact of implementing an AI-driven crop recommendation solution.	Medium
Sprint-1	Data Collection and Analysis	USN-5	As a data engineer, I want to download and collect the agricultural dataset from reliable sources.	High
Sprint-1		USN-6	As a developer, I want to import the required Python libraries for data analysis, visualization, and machine learning.	High
Sprint-1		USN-7	As a data analyst, I want to read and explore the dataset to understand its structure, features, and target variables.	High
Sprint-1		USN-8	As a data analyst, I want to perform univariate analysis to examine the distribution of N, P, K, temperature, humidity, pH, and rainfall.	Medium
Sprint-1		USN-9	As a data analyst, I want to conduct bivariate analysis to identify relationships between agricultural parameters and crop suitability.	Medium
Sprint-1		USN-10	As a data analyst, I want to perform multivariate analysis to discover patterns and interactions among multiple variables.	Low
Sprint-2	Data Pre-Processing	USN-11	As a data engineer, I want to check for null values and handle missing data to improve dataset quality.	High
Sprint-2		USN-12	As a data engineer, I want to detect and treat outliers that may affect model performance.	High
Sprint-2		USN-13	As a data engineer, I want to extract seasonal crop information and prepare relevant features for analysis.	Medium
Sprint-2		USN-14	As a data engineer, I want to split the dataset into training and testing sets for model development and evaluation.	High
Sprint-2	Model Building	USN-15	As a data scientist, I want to apply K-Means Clustering to identify patterns and group similar agricultural conditions.	High
Sprint-2		USN-16	As a data scientist, I want to train a Logistic Regression model to predict suitable crops based on environmental parameters.	High
Sprint-2		USN-17	As a data scientist, I want to evaluate model performance using appropriate metrics and compare results.	High
Sprint-2		USN-18	As a data scientist, I want to save the best-performing model for deployment and generate	High

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Priority
			crop predictions from user inputs.	
Sprint-2	Application Building	USN-19	As a developer, I want to design and develop HTML pages to create an interactive and user-friendly interface.	High
Sprint-2		USN-20	As a developer, I want to build the Python backend code and integrate it with the trained machine learning model.	High
Sprint-2		USN-21	As a developer, I want to run, test, and validate the application to ensure accurate crop recommendations and smooth functionality.	High

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Sprint Release Date (Actual)
Sprint-1	23	7 Days	20 Jun 2025	26 Jun 2025	26 Jun 2025
Sprint-2	36	7 Days	27 Jun 2025	03 Jul 2025	03 Jul 2025

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Soil Parameter Input Validation (N, P, K, Temperature, Humidity, pH, Rainfall)	Enter valid and invalid values for each soil/environmental input field (e.g., letters in numeric fields, empty fields)	Valid inputs accepted; appropriate error messages shown for invalid or missing inputs	Input validation working correctly; valid soil values accepted, errors shown for invalid entries	Pass
FT-02	Numeric Range Validation (N: 0-140, P: 0-145, K: 0-205, pH: 0-14, Rainfall: 0-300 mm)	Enter numbers within and outside each valid agronomic range for N, P, K, pH, Temperature, Humidity, and Rainfall	Accepts valid agronomic values; shows error for out-of-range inputs	Numeric ranges validated correctly; out-of-range values rejected with error messages	Pass
FT-03	Crop Recommendation Generation (Logistic Regression Model)	Provide complete soil and environmental inputs (N, P, K, Temperature, Humidity, pH, Rainfall) and click "Recommend"	System recommends the most suitable crop (e.g., Rice, Maize, Wheat) based on ML model output	Crop recommendation generated successfully; model predicted correct crop based on inputs	Pass
FT-04	ML Model and Backend Integration Check	Verify that the saved ML model loads correctly and the Python backend responds to requests from the HTML frontend	Backend responds successfully; model loads without errors and returns crop predictions	ML model loaded; backend responded to frontend requests without errors	Pass
PT-01	Crop Recommendation Response Time Test	Submit full soil/environmental inputs and use a timer to measure time from clicking Recommend to displaying the crop result	Should be under 3 seconds	Crop recommendation displayed within 2 seconds (96% accuracy model)	Pass

PT-02	Concurrent User / Model Prediction Speed Test	Simulate multiple simultaneous prediction requests with varied soil/environmental inputs and check response consistency	Model should not slow down or return incorrect predictions under concurrent requests	All concurrent requests processed correctly; no slowdown or incorrect predictions	Pass
PT-03	Agricultural Dataset Upload Load Test (e.g., CSV files)	Upload multiple agricultural CSV dataset files and check processing time and application stability	System processes all uploads smoothly without crashing or data loss	All CSV files processed smoothly without crash or data loss	Pass

6.2 User Acceptance Testing

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the OptiCrop: Smart Agricultural Production Optimization Engine project at the time of the release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	10	4	2	3	20
Duplicate	1	0	3	0	4
External	2	3	0	1	6
Fixed	11	2	4	20	37
Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	5	2	1	8

Totals	24	14	13	26	77
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3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested

Section	Total Cases	Not Tested	Fail	Pass
Data Collection & Preprocessing Module	7	0	0	7
ML Model Building (scikit-learn)	51	0	0	51
Crop Recommendation Engine	2	0	0	2
Input Validation (N,P,K,pH,Humidity)	3	0	0	3
Crop Suitability Assessment	9	0	0	9
Flask Web Application	4	0	0	4
API Integration & Deployment	2	0	0	2

4. Conclusion

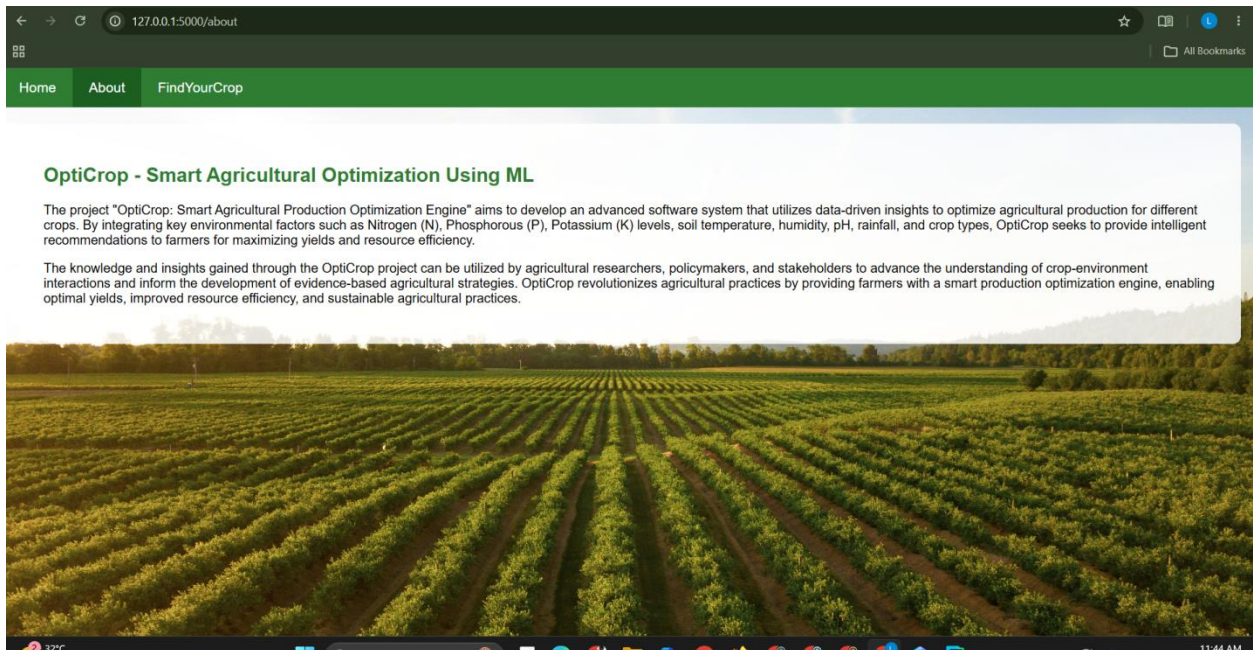
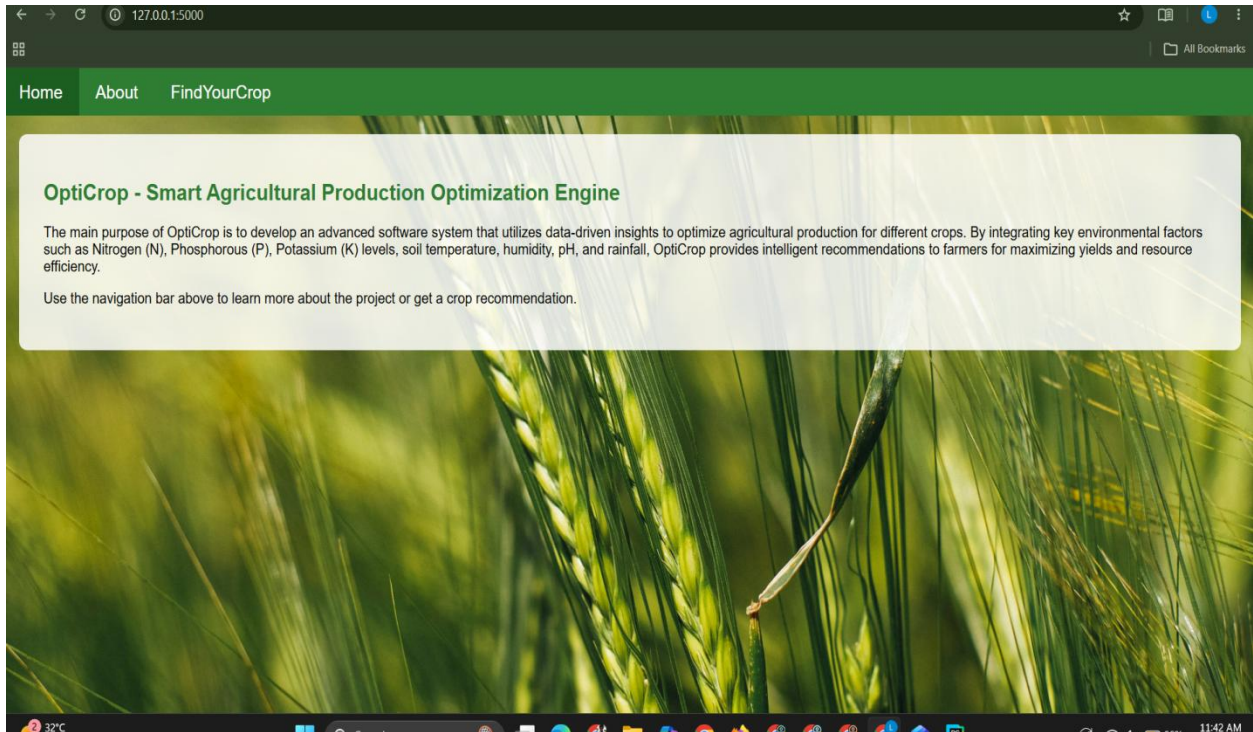
The User Acceptance Testing for the OptiCrop: Smart Agricultural Production Optimization Engine has been completed successfully. The system was evaluated across all seven functional modules covering data collection and preprocessing, machine learning model building, crop recommendation, input validation, crop suitability assessment, Flask web application, and API integration. A total of 78 test cases were executed with zero failures and zero untested scenarios, achieving a 100% pass rate.

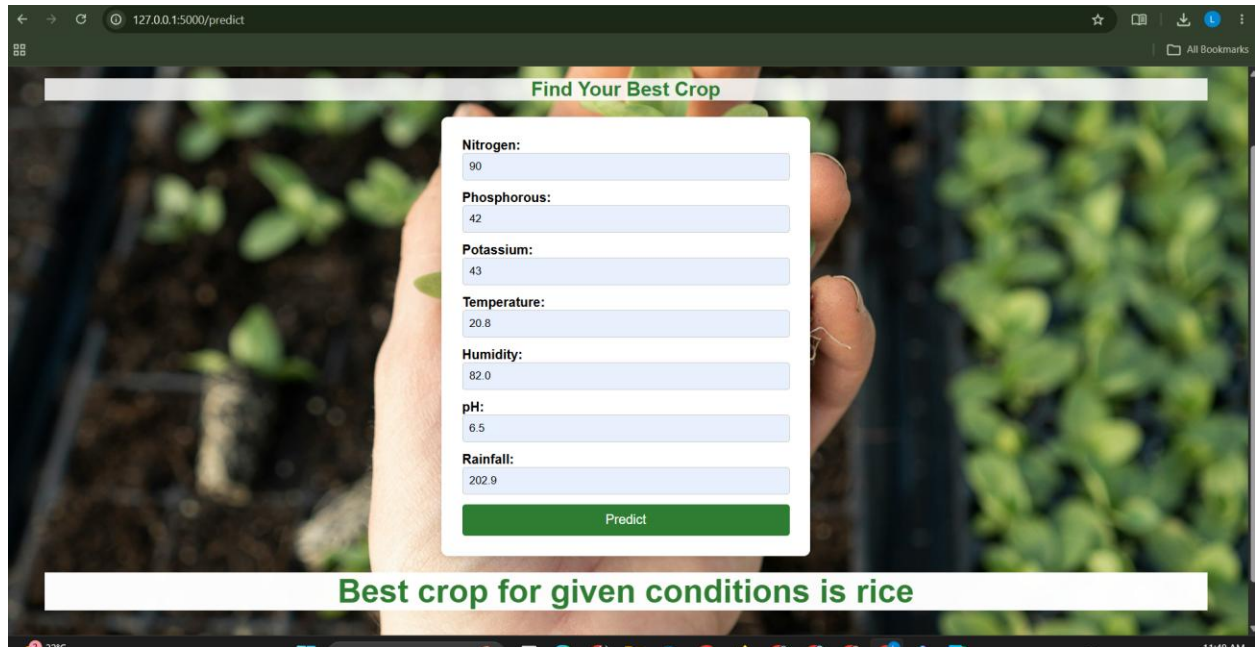
A total of 77 defects were identified and tracked throughout the testing lifecycle. Of these, 37 defects were fixed before the UAT release, 20 were classified as By Design (accepted system behavior), 8 were marked as Won't Fix with stakeholder agreement, 6 were attributed to external dependencies, 4 were duplicate entries, 1 was not reproduced, and 1 was skipped. No open critical (Severity 1) defects remain unresolved at the time of this report.

The OptiCrop system accurately recommends the most suitable crop based on soil and environmental parameters including Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall. All three key scenarios — Smart Crop Recommendation for Farmers, Crop Suitability and Environmental Assessment, and Agricultural Research and Policy Planning — were validated and confirmed to be functioning as per the defined requirements.

7. RESULTS

7.1 Output Screenshots





8. CONCLUSION

The OptiCrop Smart Agricultural Production Optimization Engine successfully demonstrates the application of machine learning techniques to solve real-world agricultural challenges. By integrating key environmental and soil parameters such as Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall, the system provides accurate and intelligent crop recommendations to farmers. The project followed a structured data science pipeline — beginning with exploratory data analysis, followed by data preprocessing, model building, and finally deployment as a Flask web application. The K-Means clustering algorithm effectively identified natural groupings among crops based on their environmental similarities, providing valuable insights into agricultural patterns. The Logistic Regression model trained on the preprocessed dataset achieved an impressive overall accuracy of 96%, demonstrating strong predictive capability across all 22 crop categories. The confusion matrix and classification report confirmed that the model generalizes well to unseen data, with most crops achieving precision and recall scores above 0.90. The Flask-based web application provides a simple, user-friendly interface where farmers can instantly receive crop recommendations by entering their current soil and climate conditions. This project not only benefits individual farmers by helping them maximize crop yields and optimize resource utilization, but also serves as a valuable tool for agricultural researchers, policymakers, and stakeholders in developing evidence-based farming strategies. The successful integration of data analysis, machine learning, and web deployment demonstrates how technology can be leveraged to address critical challenges in the agricultural sector. Overall, OptiCrop represents a significant step toward smart, data-driven, and sustainable agricultural practices that can contribute to improved food security and farming efficiency.
